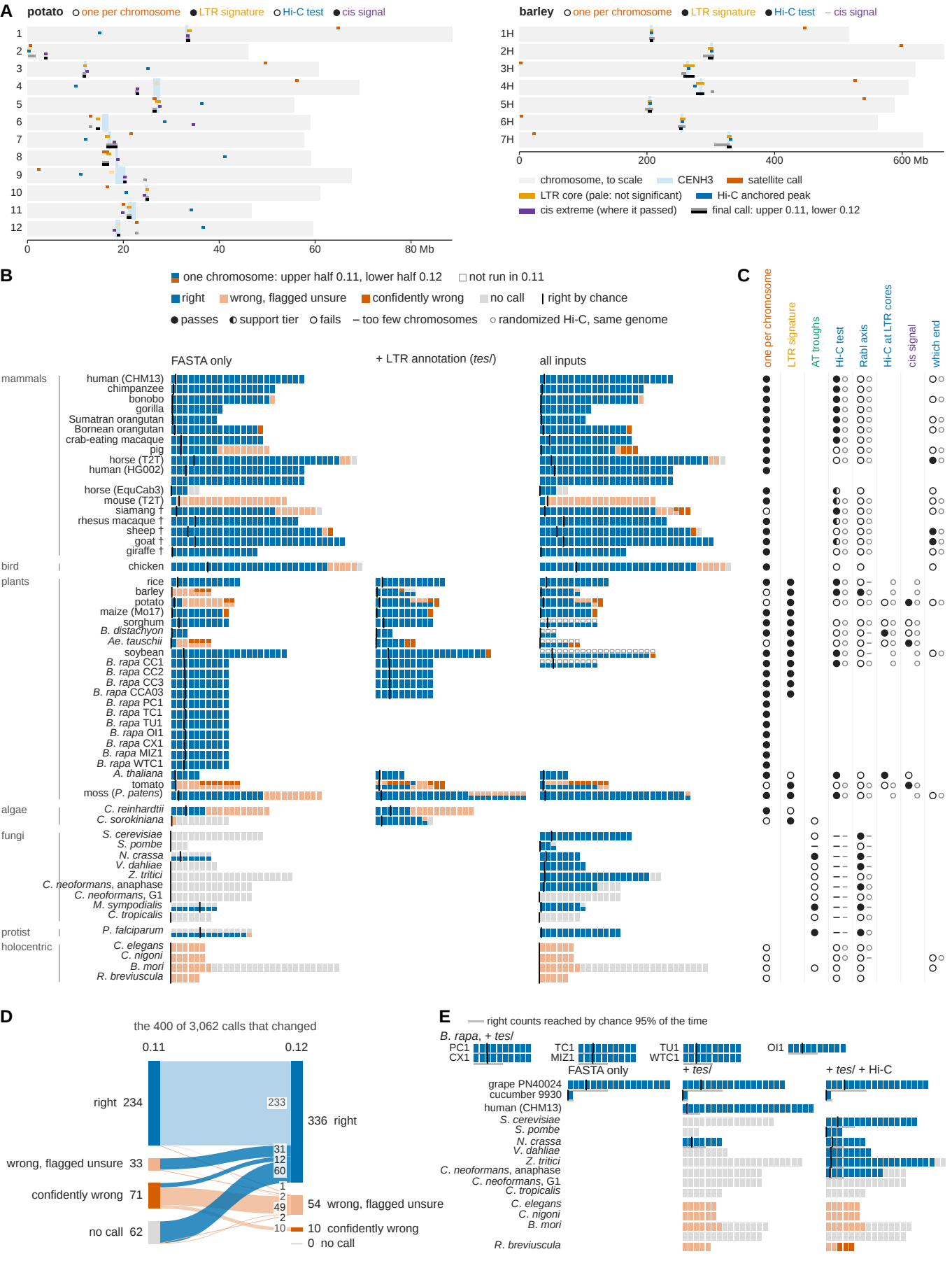

A
potato
one per chromosome
LTR signature
Hi-C test
cis signal
barley
chromosome, to scale
CENH3
satellite call
LTR core (pale: not significant)
Hi-C anchored peak
cis extreme (where it passed)
final call: upper 0.11, lower 0.12
B
one chromosome: upper half 0.11, lower half 0.12
not run in 0.11
right
wrong, flagged unsure
confidently wrong
no call
right by chance
passes
support tier
fails
too few chromosomes
randomized Hi-C, same genome
FASTA only
+ LTR annotation (tesl)
all inputs
mammals
bird
plants
algae
fungi
protist
holocentric
C
one per chromosome
LTR signature
AT troughs
Hi-C test
Rabl axis
Hi-C at LTR cores
cis signal
which end
D
the 400 of 3,062 calls that changed
0.11
0.12
right 234
wrong, flagged unsure 33
confidently wrong 71
no call 62
336 right
54 wrong, flagged unsure
10 confidently wrong
0 no call
E
right counts reached by chance 95% of the time
B. rapa, + tes/
FASTA only
+ tes/
+ tes/ + Hi-C

When each line of evidence must pass a test in the genome at hand before it may act, more centromere calls are right and half as many are confidently wrong. Evidence that is wrong across a whole genome fails its test and stays silent.

A Potato and barley, every chromosome to scale: each line's own centromere call, the final call of each release and CENH3. Satellites point elsewhere and fail their test; the cores of one retrotransposon family and the contact signals sit on CENH3, and the final call follows them.

B Every chromosome of 56 genomes, from apes and crops to yeast and the malaria parasite, before (0.11) and after (0.12) the tests were added: confident errors in tomato, potato, barley and *Ae*. *tauschii* become right or flagged calls, and AT-rich centromeres of fungi and *Plasmodium* are now found from sequence alone.

C Which tests pass depends on the kind of centromere: satellites fall once per chromosome in mammals and most plants, one retrotransposon family marks plant centromeres, AT troughs mark fungi and *Plasmodium*, and contact maps add the rest. Randomized contact maps pass nothing.

D Every call that changed between the releases: confident errors drain into right or flagged calls, chromosomes without a call gain right ones, and no call became confidently wrong.

E Genomes and annotations kept out of the rule-making behave the same way: right far beyond chance, with no confident error.

Terms. A call is right when it lies on the mapped centromere (CENP-A or CENH3 ChIP, CUT&Tag, or published coordinates), within each benchmark's tolerance (overlap in mammals; 1 Mb or 4 Hi-C bins in plants; 1 Hi-C bin in fungi). Each call is *high*, *medium* or *low*: a wrong *high* or *medium* call is confidently wrong, a wrong *low* call is flagged unsure; in holocentric genomes any call is wrong. Chance: a chromosome's chance of a right call by luck is (call + centromere + 2 × tolerance) / chromosome length; summed, the right calls expected by luck. Both releases ran on the same inputs, scored by the same rules. Over all 218 scored runs (56 genomes; with the 2 *Poa annua* runs, which have no truth, the 220 release runs) confident errors fell from 104 to 52.

A The raw evidence of the claim on every chromosome of two genomes: potato (12) and barley (7), each a gray lane to its own bp scale; light blue: CENH3 (none on potato chromosome 2). In rows, each line's own locus: the satellite call (FASTA alone), the core of the LTR signature family (pale: not significant on that chromosome), Hi-C's anchored peak, the cis extreme (where that test passed); beneath, the final call with all inputs, 0.11 (gray) above 0.12 (black). On CENH3 (within the scoring tolerance, potato 1 Mb, barley 8 Mb) in potato: satellite calls 2 of 11, LTR cores 10 of 11, Hi-C peaks 0 of 11, cis extremes 9 of 11, final calls 6 of 11 (0.11) and 9 of 11 (0.12); in barley: satellite calls 0 of 7, LTR cores 7 of 7, Hi-C peaks 6 of 7, final calls 6 of 7 and 7 of 7. In both, the satellites fail the one-per-chromosome test and so do not act; the LTR signature decides; Hi-C, tested against those calls, fails in potato (its peaks scatter), where the cis signal acts instead, and passes in barley. Selection: of the genomes where the satellite fails its test, the LTR signature decides and Hi-C is tested (barley, potato, *Ae*. *tauschii*, tomato), potato and barley show both outcomes of the Hi-C test (tomato is the subject of Fig. 1). Every chromosome of both is drawn, misses included.

B One cell per chromosome with a mapped centromere, upper half 0.11, lower half 0.12 (1,663 cells), sorted by the 0.12 outcome; black tick: right calls expected by luck. Columns: FASTA only (56 genomes); with the LTR retrotransposon annotation (*tes*/, 17); the richest input set where it adds to these (44: gene models and Hi-C in mammals, Hi-C in fungi and holocentric genomes, *tes*/ with Hi-C or methylation in plants, methylation in HG002). Confident errors fell from 19 to 3 with FASTA only (right 526 to 554), 14 to 7 with *tes*/ (right 144 to 167) and 18 to 9 with all inputs (right 485 to 501, on the 623 chromosomes run in both; open: 50 chromosomes whose *tes*/ + Hi-C set is new in 0.12). Over the 47 genomes with functional truth, each with all its inputs, 550 of 633 chromosomes are right and 9 confidently wrong. † no CENP-A data: scored against the satellome locus of another caller.

C The tests a line must pass before it may act, as run on each genome (filled: passes; half: Hi-C reaches only the support tier; open: fails; dash: too few chromosomes to test; blank: not run, its input not given or nothing to test). One per chromosome: the N largest satellite clusters cover more chromosomes than chance (36 of 46 genomes with satellites). LTR signature: one retrotransposon family is enriched at a core on most chromosomes and decides (15 of 17). AT troughs: the deepest GC troughs fall once per chromosome (3 of 11). Hi-C test: contact peaks agree with the other chromosomes' calls (14 of 32; support tier 4). Rabl axis: the arms fold at, or align by distance from, the centromeres (7 of 42). Hi-C at LTR cores and cis signal: run where *tes*/ is given and the Hi-C test fails (2 and 3 of 6). Which end: run where a call sits at a chromosome end (3 of 18). Gray: the same tests on randomized Hi-C of the same genome (pairs re-paired, or ends moved; 57 runs, 37 genomes): none passes, and none of the 56 runs with a partner differs in any call from it. The sequence tests carry their own nulls: shuffled LTR annotations gave a signature 0 times in 157; GC blocks shuffled among chromosomes passed the AT test 16 times in 360, as its 5% level allows.

D The 400 of 3,062 calls (the 202 runs scored in both releases) whose outcome, place or confidence changed; band widths are chromosomes; 189 moved. 49 confident errors became flagged (tomato 17, *Ae*. *tauschii* 12, siamang 11, potato 6, barley 3: satellites that fail the one-per-chromosome test, and the cis signal); 12 became right (tomato 9, potato 2, barley 1, with *tes*/). 60 chromosomes without a call gained a right one (*P*. *falciparum* 26, *M*. *sympodialis* 17, *N*. *crassa* 14, barley 2, *S*. *pombe* 1; AT troughs; 28 of them in the randomized-Hi-C runs, which equal the FASTA-only runs) and 2 a flagged wrong one (*P*. *falciparum* 2). 31 flagged errors became right (potato 12, moss (*P*. *patens*) 11, tomato 6, barley 1, *C*. *sorokiniana* 1: the LTR signature). 1 right call became a flagged error (tomato 1, *tes*/ alone). In all, right calls rose by 102 (2,330 to 2,432) and confident errors fell by 61 (104 to 43); 0 calls became confidently wrong.

E Arms scored only after the 0.12 rules were frozen: LTR annotations made for this release on genomes never run with them (the seven held-out *B*. *rapa* accessions as a block), and two genomes new to the benchmark (grape PN40024, cucumber 9930, CENH3 from their authors), drawn as in B (0.12 only). The panel is laid out from the 38 planned arms; 38 were scored when this figure was drawn and the others are hatched (not finished in time). Scored: 38 arms of 22 genomes, 422 chromosomes: 220 right where 35.5 are expected by luck, 3 confidently wrong; 18 of the 21 arms with calls are right beyond chance ($P < 0.01$, Poisson-binomial; cucumber has truth on one chromosome). With *tes*/ alone fungi get no call (no LTR family marks their centromeres); with Hi-C they do. Holocentric genomes: 3 confident calls in 8 arms (all in *R*. *breviuscula* with Hi-C, whose LTR cores lie at chromosome ends, which cluster in its nuclei; under the rule change they prompted, those calls are *low*). Gray bar: the right counts luck reaches 95% of the time. The 2 FASTA + Hi-C arms of grape and cucumber are not drawn, for space (2 scored).

Caveats. The 17 genomes with *tes*/ were the development set of the 0.12 rules; E tests that. 9 rules of 0.12 were changed after a result was seen, each for a reason independent of that case and rechecked on others (cenfuse benchmark, section 12): the one-per-chromosome cap, the 10-fold core floor, thin decisive cores, weak cores not rejecting families, Hi-C partner windows kept at the LTR cores, the observed/expected cis profile, its tolerance, AT troughs placing only what nothing else places, and LTR-core peaks at chromosome ends not counting (after held-out *R*. *breviuscula*). Mouse and chicken are unchanged (mouse stays 1 of 20 right, its other 19 calls flagged unsure). Confidence is not split on the figure; with all inputs, *high* calls are right 406 of 410 times, *medium* 62 of 67, *low* 82 of 124. No single example is chosen: A draws every chromosome of the two genomes picked by the rule above, B every chromosome of every genome.

Analysis: asc-oulab:/data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/v10.3/v10.4/v11 · Script: /data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/v10.3/v10.4/v11/figures/fig_integration/fig_integration.py · 2026-11-07